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Course: Data Science

LOANGUARD AI

Loan Approval System

Final Project Report · Version 1.0

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1. Executive Summary

LoanGuard AI is a machine learning-powered credit risk prediction system designed to assist financial institutions in making fast, accurate, and data-driven loan approval decisions. The system analyses an applicant's financial profile across 20 key variables and predicts whether they are likely to fully repay a loan or default on it.

After evaluating five baseline algorithms and two advanced ensemble approaches, XGBoost (Extreme Gradient Boosting) was selected as the final production model due to its superior predictive performance. The trained model is deployed through a Streamlit web interface that enables real-time risk assessment without requiring any technical expertise from the end user.

2. Project Introduction

2.1 Background

Loan default is one of the most significant financial risks faced by banks and lending institutions. Traditional credit evaluation methods are often slow, inconsistent, and dependent on a limited number of variables. Machine learning offers a scalable and objective alternative — one capable of analysing dozens of financial signals simultaneously to produce a reliable risk verdict.

2.2 Objectives

- Build a machine learning model that accurately classifies loan applicants as likely to repay (Fully Paid) or likely to default (Charged Off).
- Engineer meaningful financial ratios from raw data to improve model insight.
- Deploy the solution as a user-friendly web application accessible to non-technical users.
- Apply industry best practices including class balancing, feature selection, and hyperparameter optimisation.

2.3 Project Structure

The project consists of two primary files. The Jupyter Notebook (Loan_Approval_System.ipynb) contains the full data science pipeline including EDA, preprocessing, feature engineering, model training, and evaluation. The Streamlit application (app.py) serves as the production-facing interface, loading the saved model to deliver real-time predictions.

3. Dataset Overview

The dataset was sourced from `Loan_Approval_Dataset.csv`, containing historical credit records of loan applicants. After removing 514 entirely empty trailing rows, the final usable dataset was obtained. The target variable distinguishes between Fully Paid (1 — approved) and Charged Off (0 — rejected) loans. Input features span three categories: Applicant Profile (Annual Income, Employment Tenure, Home Ownership), Loan Details (Amount, Term, Purpose, Monthly Debt), and Credit History (Credit Score, Open Accounts, Bankruptcies, Tax Liens, and more).

The dataset exhibited a class imbalance with approved loans outnumbering defaults. This was addressed during preprocessing using SMOTE oversampling applied exclusively to the training set, preserving the real-world distribution in the test set.

4. Exploratory Data Analysis (EDA)

EDA was conducted to uncover patterns, distributions, and relationships within the data before modelling. Key findings are summarised below.

Loan Term vs. Default Rate: Long-term loans showed a significantly higher default rate compared to short-term loans, making loan term one of the strongest predictors in the dataset.

Employment Tenure: Applicants with 10+ years in their current job had the highest approval count, while those with less than one year showed elevated rejection rates, indicating a strong link between employment stability and repayment behaviour.

Loan Purpose: Debt consolidation was the most frequent loan purpose and exhibited relatively stable repayment behaviour. Several rare categories (e.g. wedding, vacation, moving) were grouped under “other” due to low frequency and higher default tendency.

Home Ownership: Applicants with a mortgage or who own their home outright demonstrated higher approval rates than renters, suggesting greater financial stability among homeowners.

Class Imbalance: The distribution of the target variable was imbalanced, with Fully Paid loans considerably outnumbering Charged Off cases. This imbalance was observed during EDA and informed the decision to apply SMOTE during preprocessing.

5. Data Preprocessing & Cleaning

4.1 Column Removal & Imputation

Loan ID and Customer ID were removed as unique identifiers with no predictive value. Months Since Last Delinquent was dropped due to over 50% missing values. Remaining missing values were addressed as follows: Credit Score, Annual Income, and Maximum Open Credit were filled with the median (robust to outliers); Years in Current Job was filled with the mode (categorical); Bankruptcies and Tax Liens were filled with 0 (absence implies none).

4.2 Outlier Treatment

Credit Score values were clipped to the valid real-world range of 300–850, correcting data-entry errors where scores appeared in the thousands. All remaining numerical columns were treated via the IQR method — values beyond 1.5x the interquartile range were capped at fence values rather than removed, preserving row count.

4.3 Categorical Encoding

Column	Method	Detail
Term	Label Encoding	Short Term = 1, Long Term = 0
Years in Current Job	Ordinal Mapping	< 1 year = 0 through 10+ years = 10
Home Ownership	One-Hot Encoding	Dummy trap avoided via drop_first=True
Purpose	One-Hot Encoding	7 rare categories consolidated as 'other'

6. Feature Engineering

Three financial ratio features were derived from existing variables to capture risk signals that raw figures alone cannot convey.

Feature	Formula	Risk Threshold
Debt-to-Income Ratio (DTI)	Monthly Debt / (Annual Income / 12)	> 40% indicates high risk
Credit Utilisation Ratio	Current Balance / (Max Credit + 1)	> 30% signals over-reliance on credit
Loan-to-Income Ratio (LTI)	Loan Amount / Annual Income	Lower ratio = manageable repayment burden

A Random Forest model with 300 estimators was trained solely to rank feature importance. The top 20 features by importance score were retained, reducing the feature space from 27 to 20 and improving model generalisation.

7. Model Development & Selection

6.1 Baseline Comparison

Model	Characteristics
Logistic Regression	Linear boundary; limited on non-linear patterns
Decision Tree	Interpretable but prone to overfitting without pruning
K-Nearest Neighbours	Distance-based; moderate on structured tabular data
Naive Bayes	Fast and simple; assumes full feature independence
Random Forest (Baseline)	Strong ensemble; primary baseline for comparison

6.2 Final Model — XGBoost

XGBoost was selected as the final production model after outperforming all candidates including a tuned Random Forest (GridSearchCV over 48 combinations) and a Stacking Classifier. It was tuned using RandomizedSearchCV over 20 iterations optimising for F1-score, across hyperparameters: n_estimators (200–500), learning_rate (0.01–0.1), max_depth (3–6), min_child_weight, gamma, subsample, and colsample_bytree.

6.3 Classification Threshold

Rather than the default 0.50 threshold, the final model uses 0.71. An applicant must achieve a predicted repayment probability of at least 71% before being classified as approved. This conservative threshold deliberately reduces false approvals, prioritising lender protection. The trained model and StandardScaler are serialised to disk as `xgb_model.pkl` and `scaler.pkl` via `joblib` for efficient runtime loading.

8. Application Interface

The Streamlit application (`app.py`) provides a professional, dark-themed web interface for real-time loan risk assessment. It is organised into four input panels: Applicant Profile (Annual Income, Employment Tenure, Home Ownership), Loan Request (Amount, Term, Purpose, Monthly Debt), Credit Health (Credit Score, Credit History, Open Accounts, Credit Limit), and Advanced Details — expandable panel for Current Credit Balance, Credit Problems, and Bankruptcies.

Upon clicking Execute Risk Assessment, the app constructs the feature vector, computes DTI, LTI, and Utilisation in real time, applies the same encoding and scaling used during training, passes the vector through XGBoost, and renders the verdict as a full-width card displaying either LOAN APPROVED or LOAN REJECTED alongside the three supporting financial ratios.

9. Loan Approval & Rejection Criteria

8.1 Conditions Favouring Approval

- Low DTI (below 40%) and low credit utilisation (below 30%), indicating comfortable repayment capacity.
- Low Loan-to-Income ratio — loan amount is proportionate to annual earnings.
- Good credit score (300–850 range) with a long credit history and zero bankruptcies.
- Short-term loan selection — EDA confirmed significantly lower default rates vs. long-term.
- Stable employment of 10+ years and home ownership (mortgage or owned property).
- Debt consolidation as the loan purpose — most common and most stable category in the dataset.

8.2 Conditions Leading to Rejection

- High DTI, high credit utilisation, or high LTI indicating over-extension of finances.
- History of bankruptcies or recorded public credit problems.
- Long-term loan selection — disproportionately higher default rate found in EDA.
- Short employment tenure (less than one year) signalling income instability.
- Predicted repayment probability below the 71% approval threshold.

10. Conclusion

LoanGuard AI successfully demonstrates the end-to-end development of a credit risk prediction system — from raw data through to a deployed, interactive application. The project addresses real-world challenges including missing data, class imbalance, high-dimensional features, and the need for interpretable outputs for non-technical users.

The selection of XGBoost as the final model, combined with deliberate feature engineering (DTI, LTI, Credit Utilisation), SMOTE-based class balancing, and a conservative classification threshold of 0.71, results in a system that prioritises precision in risk identification. Future enhancements could include SHAP-based feature explanation for individual predictions, probability score visualisation, multi-applicant batch processing, and integration with live credit bureau APIs for automated data retrieval.